Setting Up Your Deep Learning Environment with Anaconda

This document walks you through in setting up your Deep Learning Environment by downloading and installing Anaconda, updating scikit-learn library, installing various Deep Learning libraries and optionally installing and setting up your Integrated Development Environment, PyCharm Community Edition.

1. Download Anaconda
   a. Visit https://www.anaconda.com/products/individual#Downloads and select the installer for your platform (Linux, Windows, etc.)
   b. Follow installation instructions for your platform at https://docs.anaconda.com/anaconda/install/ to perform installation. 'Thank you for installing Anaconda !' message completes your Anaconda installation.
   c. Open a terminal on Linux or a command line window on Windows and type the following command to confirm that installation has been correctly completed.
      - **`conda -V`**
      - which should return the version number of conda, such as **`conda 4.10.1`** (or something similar)
   d. Confirm that Python has been installed correctly by typing
      - **`Python -V`**
      - which should return the version number of Python, such as **`Python 3.7.4`** (or something similar)
   e. Confirm that your conda environment is up-to-date by typing
      - **`conda update conda`**
      - **`conda update anaconda`**
      - You may need to install some packages and confirm the updates.
   f. Confirm that your SciPy, NumPy, Matplotlib, Pandas, Statsmodels, and Scikit-learn is up-to-date by utilizing the script below:

```
# scipy
import scipy
print('scipy: %s' % scipy.__version__)
# numpy
import numpy
print('numpy: %s' % numpy.__version__)
# matplotlib
import matplotlib
print('matplotlib: %s' % matplotlib.__version__)
# pandas
import pandas
print('pandas: %s' % pandas.__version__)
# statsmodels
import statsmodels
print('statsmodels: %s' % statsmodels.__version__)
# scikit-learn
import sklearn
print('sklearn: %s' % sklearn.__version__)
```

- You can save the above script to **`versions.py`** file and then run it by typing **`python versions.py`** on terminal.
- which should return the version numbers as follows (or something similar)

```
scipy: 1.3.2
numpy: 1.16.4
matplotlib: 3.1.1
pandas: 0.25.3
statsmodels: 0.10.1
sklearn: 0.22.1
```

2. Update scikit-learn library
   a. Type the following command on terminal to update scikit-learn library
      - **`conda update scikit-learn`**
   b. Confirm the update by re-running **`versions.py`** script
3. Install Deep Learning libraries. This document uses Keras for Deep Learning library, which requires one of Theano or TensorFlow to be installed. Please use one of Theano or TensorFlow, not both.
   a. To install Theano Deep Learning library type the following command on terminal
      - **`conda install Theano`**
   b. To install TensorFlow Deep Learning library type the following command
      - **`conda install -c conda-forge tensorflow`**
   c. Install Keras by typing the following command
      - **`pip install keras`**
   d. Confirm that Deep Learning libraries have been installed correctly by utilizing the script below:

```
# theano
import theano
print('theano: %s' % theano.__version__)
# tensorflow
import tensorflow
print('tensorflow: %s' % tensorflow.__version__)
# keras
import keras
print('keras: %s' % keras.__version__)
```

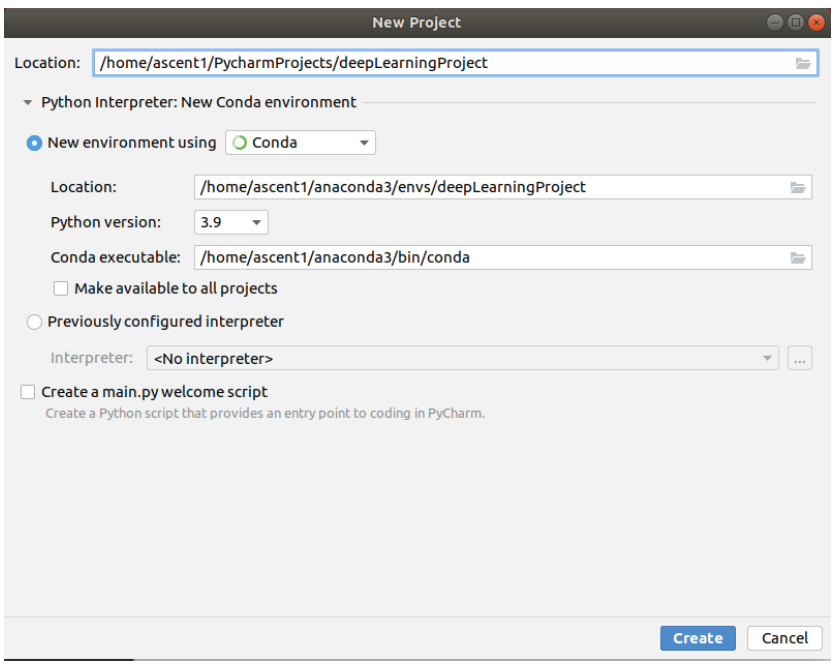
- You can save the above script to **`dl_versions.py`** file and then run it by typing **`python dl_versions.py`** on terminal.
- which should return the version numbers as follows (or something similar)

```
tensorflow: 1.14.0
Using TensorFlow backend.
keras: 2.3.1
```

This concludes the installation of Anaconda, various libraries and Deep Learning frameworks.

Optional: Install PyCharm Community Edition

1. Download PyCharm Integrated Development Environment
    a. Visit https://www.jetbrains.com/pycharm/download/#section=linux and select the installer for your platform (Linux, Windows, etc.) Make sure you download Community (not Professional)
    b. Copy the pycharm-2021.1.1.tar.gz (for Linux) to the desired installation location, update the folder where the installer is moved over by ensuring that it has **`rw`** permissions
    c. Unpack the installer to an empty directory by typing the following command on terminal
        - **`tar -xzf pycharm-2021.1.1.tar.gz`**
    d. Run PyCharm by typing the following command
        - **`pycharm-community`**
        - This will bring up **`Welcome to PyCharm`** window, in which click on **`New Project`**
        - This will bring up **`New Project`** window, where you enter a project name, **`deepLearningProject`** and click on **`create`**. Ensure that **`Conda`** is selected for your environment as follows



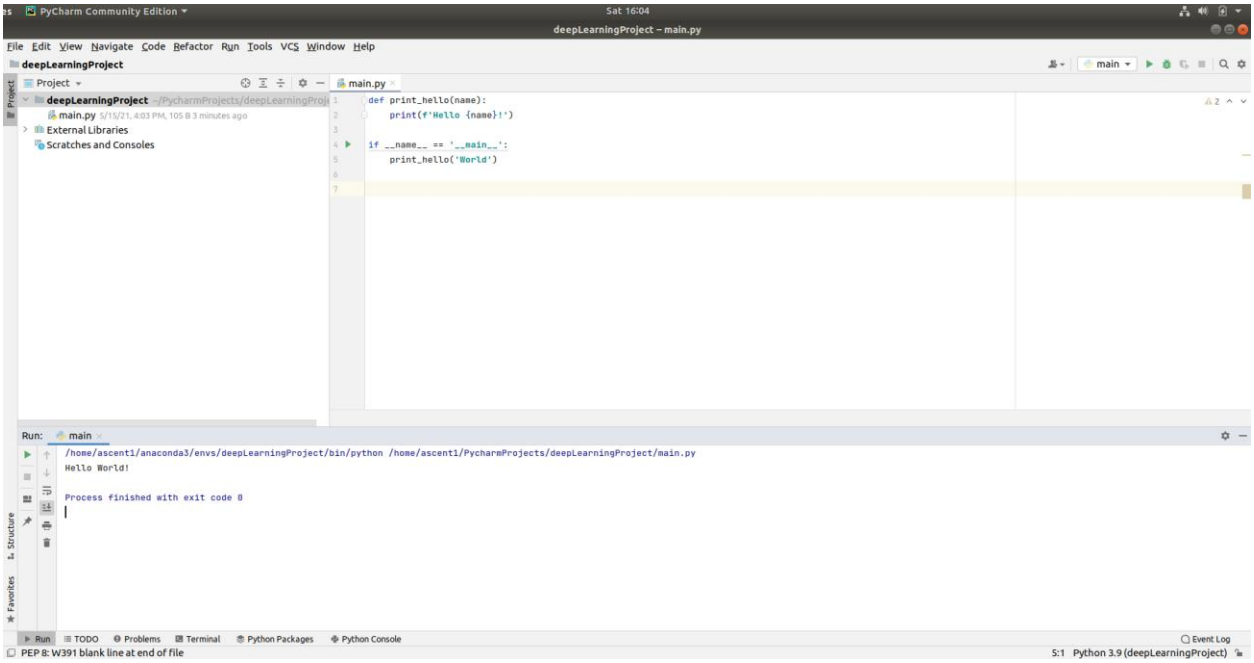
    e. On the left in the Project window, select **`deepLearningProject`** and right mouse click to create a new Python file, main.py
    f. This will create **`main.py`**. Copy the content below to your main.py file

```
def print_hello(name):
    print(f'Hello {name}!')

if __name__ == '__main__':
    print_hello('World')
```

g. Run **`main.py`** file by clicking on the green arrow button on the upper right corner.

- This will print **`'Hello World!'`** as follows:



This concludes the optional installation of PyCharm Community Edition.